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ONCOLOGY

Juxtapapillary and circumpapillary choroidal melanoma: globe-sparing treatment outcomes with iodine-125 notched plaque brachytherapy

John V. Hegde[1] · **Tara A. McCannel**[2] · **Colin A. McCannel**[2] · **James Lamb**[1] · **Pin-Chieh Wang**[1] · **Darlene Veruttipong**[1] · **Robert Almanzor**[2] · **D. Jeffrey Demanes**[1] · **Mitchell Kamrava**[1]

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Abstract

Purpose Managing juxtapapillary and circumpapillary choroidal melanoma with brachytherapy is challenging because of technical complications with accurate plaque placement and high radiation toxicity given tumor proximity to the optic nerve. We evaluated our center's experience using ultrasound-guided, Iodine (I)-125 notched plaque brachytherapy for treating choroidal melanoma contiguous with (juxtapapillary) and at least partially surrounding the optic disc (circumpapillary).

Methods All cases of choroidal melanoma treated with I-125 notched plaque brachytherapy at our center from September 2003–December 2013 were retrospectively reviewed. Only patients with ≥18 months of follow-up who had lesions contiguous with the optic disc (0 mm of separation) were included. The tumor apex prescription dose was 85 Gy. Outcomes evaluated included local control, distant metastasis-free survival (DMFS), cancer-specific survival (CSS), overall survival (OS), visual acuity, and radiation toxicity.

Results Thirty-four patients were included with a median follow-up of 44.1 months (range 18.2–129.0). AJCC T-category was T1 in 58.8%, T2 in 26.5%, and T3 in 14.7%. Median circumferential optic disc involvement was 50% (range 10%–100%). Eye retention was achieved in 94.1%. Actuarial 2- and 4-year rates of local recurrence were 3.1% and 7.6%, DMFS were 97.0% and 88.5%, CSS were 97.0% and 92.8%, and OS were 97.0% and 88.9%, respectively. In addition, 23.5% had visual acuity ≥20/200 at last follow-up.

Conclusions I-125 notched plaque brachytherapy provides high eye preservation rates with acceptable longer-term post-treatment visual outcomes. Based on our experience, choroidal melanoma directly contiguous with and partially encasing the optic disc may be effectively treated with this technique.

Keywords Choroidal melanoma · Circumpapillary · Juxtapapillary · Ultrasound guidance · Notched plaque · Brachytherapy

Introduction

The long-term results of the Collaborative Ocular Melanoma Study (COMS) for medium-sized choroidal melanomas established Iodine (I)-125 brachytherapy as a primary treatment option by demonstrating no difference in overall survival between I-125 brachytherapy and enucleation for lesions >2 mm from the optic disc [1].

For lesions either <2 mm from the optic disc (juxtapapillary) or those wrapping around the optic disc (circumpapillary), there are multiple options for management, including various plaque brachytherapy techniques [2, 3]. Improvements in plaque design, using either a notched or slotted technique, in addition to the utilization of ultrasound guidance for confirmation of appropriate plaque placement, can be employed to overcome the perceived limitations of brachytherapy for these more challenging-to-treat lesions. Adjuvant transpupillary

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✉ John V. Hegde
jhegde@mednet.ucla.edu

[1] Department of Radiation Oncology, 200 UCLA Medical Plaza, Suite B265, University of California, Los Angeles, Los Angeles, CA 90095, USA

[2] Department of Ophthalmology, Stein Eye Institute, 100 Stein Plaza UCLA, University of California, Los Angeles, Los Angeles, CA 90095, USA

thermotherapy has also been utilized. While local control outcomes in peripapillary lesions treated with notched/slotted brachytherapy are encouraging, data for juxtapapillary lesions contiguous with the optic disc and circumpapillary lesions which encircle the optic disc are more limited [2, 3]. We report on our 10-year clinical experience with notched plaque brachytherapy for juxtapapillary choroidal melanoma, which at least partially encircles the optic disc and has 0 mm of separation between the tumor and the disc.

Methods

A retrospective, Institutional Review Board-approved database review of patients treated with I-125 notched plaque brachytherapy for previously untreated juxtapapillary and circumpapillary choroidal melanoma from September 2003 through December 2013 at the University of California, Los Angeles (UCLA) Medical Center was performed. For this type of study, formal consent is not required. Inclusion criteria included patients with choroidal melanoma directly contiguous with the optic disc, with no margin (0 mm) separating the tumor and the disc, and with all lesions at least partially wrapping around the disc. Patients with less than 18 months of follow-up were excluded from analysis. No adjuvant therapy was performed following notched plaque brachytherapy.

Patient charts were reviewed for baseline clinical characteristics, including age, gender, vision-specific comorbidities, and baseline visual acuity as measured by Snellen best-recorded visual acuity and by the logarithm of the minimum angle of resolution (LogMAR) scale. The following tumor characteristics were also recorded: distance from the optic disc, tumor apical height, and tumor greatest basal diameter. Distance from the optic disc and percent of the optic disc encircled by tumor were measured and recorded by the treating ophthalmic oncologist. Tumor height and basal diameter were measured by ultrasound.

I-125 notched plaque brachytherapy was performed on all patients by an ophthalmic oncologist, who also performed a biopsy for diagnostic confirmation during plaque placement. The plaques were engineered by our institution's physics faculty within the department of radiation oncology. The plaque shells consisted of a molded 1/100th inch-thick gold foil with integrated anterior suture eyelets. Hemi-circular notches were cut out of the shells on the opposite end from the suture eyelets. I-125 seeds were affixed with cyanoacrylate in predetermined patterns specific to each plaque size. They were then covered by a layer of polymethyl methacrylate (PMMA), with the interior surface conformed to a globe of radius 25 mm. Radiation dose was calculated per the COMS protocol using the American Association of Physicists in Medicine TG-43 U1 methodology [4, 5], as TG-129 methodology was available only toward the conclusion of the interval within which this cohort was treated [6]. Because the plaques used nearly water-equivalent molded PMMA instead of the silicone elastomer insert used by COMS plaques, the plaques were not subject to the dose reduction of approximately 10% due to the silicone elastomer transmission factor that has been reported for COMS plaques [5].

Dose rates were within the acceptable dose rate range reported in the COMS study, approximately 50 cGy/h. Plaques were built to encompass the widest basal dimension of the tumor with 3-mm lateral margins. The prescription dose was 85 Gy to the apical height of the tumor. An ultrasound-guided plaque placement technique was used to optimize tumor-plaque apposition [7]. All patients had notched plaque placement in a similar manner such that all the tumor was covered, regardless of the degree of circumferential involvement of the tumor, prioritizing full tumor coverage over reduction of dose to the optic disc. In addition, silicone oil was used in 20.6% of patients, as a vitreous substitute to attenuate radiation to the non-tumor tissues of the eye [8]. Notably, all patients were treated prior to the release of the recent American Brachytherapy Society (ABS) guidelines for plaque brachytherapy of choroidal melanoma [9].

After brachytherapy treatment, patients were evaluated at 3-month intervals until 6 months post-treatment and then every 6 months thereafter. Patients were also evaluated by their primary physician or medical oncologist for metastatic surveillance every 6 months. Local recurrence was defined as >0.5 mm apical tumor growth with continued increases in height after at least two follow-up visits. Actuarial 2- and 4-year local tumor control, distant metastasis-free survival, cancer-specific survival, and overall survival were recorded for each patient. Actuarial probability rates with 95% confidence intervals (CI) were calculated using Kaplan-Meier survival curve analysis.

The following interval follow-up information was also recorded for all patients at each visit: visual acuity, tumor apical height, and radiation side effects (changes in visual acuity, radiation retinopathy, optic atrophy, and neovascular glaucoma). All patients had multi-modality evaluations with ultrasonography, color fundus photography, fluorescein angiography, and/or ocular coherence tomography (OCT) imaging obtained serially to monitor the patient's treatment response and toxicity evaluation. At some follow-up visits, reliable ultrasound or OCT imaging may have been unable to be reliably performed for technical reasons. Also, poor visualization of the fundus due to media opacity or tumor involvement of the macula, for example, limited visual evaluation at some follow-up visits. Therefore, limited radiation toxicity data was available for some patients in this cohort.

Results

Patient characteristics

Table 1 reports the baseline patient characteristics. Of the entire cohort ($N = 41$) having choroidal melanoma directly contiguous with the optic disc, seven patients with less than 18 months of follow-up were excluded from analysis. Therefore, N = 34 patients were available for analysis. The median follow-up duration after completing treatment was 44.1 months (range 18.2–129.0). The median age of patients at the time of diagnosis was 57.5 years (range 24.8–85.0). Thirty-eight percent of patients had hypertension, while 8.8% had diabetes mellitus at the time of diagnosis. Median tumor apical height prior to plaque placement was 3.3 mm (range 1.0–10.2). Median maximal basal tumor dimension was 8.6 mm (range 4.5–17.0). According to the American Joint Committee on Cancer (AJCC) 7th Edition, 58.8% of patients had a T1 lesion, 26.5% had a T2 lesion, and 14.7% had a T3 lesion. A median tumor wrap of 50% (encircling 180^0 of the optic disc) (range 10–100%) was noted, with 55.9% having ≥180^0 wrap around the optic disc.

Local tumor control

Actuarial 2- and 4-year rates of local tumor control were 96.9% (CI 80.1–99.6) and 92.4% (CI 72.3–98.1), respectively (Table 2, Fig. 1). Local tumor recurrence occurred in three cases, and the median time to local recurrence was 38.7 months (range 13.6–72.5). Of the three local recurrence cases, one was observed, the second underwent re-treatment with I-125 plaque brachytherapy, and the third underwent enucleation. In the enucleated specimen, residual malignant melanoma was found. No patient with local recurrence developed distant metastases, and all patients with recurrent disease remain alive at last follow-up.

Table 1 Baseline Patient Characteristics

Median age at diagnosis (range)	57.5 years (24.8–85.0)
Median ultrasound assessed apical tumor height prior to plaque placement (range)	3.3 mm (1.0–10.2)
Median maximum ultrasound assessed basal tumor dimension (range)	8.6 mm (4.5–17.0)
Median plaque diameter (range)	15.5 mm (11–23)
AJCC T-Category	
T1	20/34 (58.8%)
T2	9/34 (26.5%)
T3	5/34 (14.7%)
T4	0/34 (0)
Median percent tumor wrap (range)	50.0% (10–100%)
Percent tumor wrap category	
< 25%	3/34 (8.8%)
25% to < 50%	12/34 (35.3%)
≥ 50 to < 75%	18/34 (52.9%)
≥ 75%	1/34 (2.9%)
Median patient follow-up (range)	44.1 months (18.2–129.0)

Choroidal melanoma distant metastasis & survival

Actuarial 2- and 4-year rates of distant metastasis-free survival were 97.0% (CI 80.4–99.6) and 88.5% (CI 68.1–96.2), respectively (Table 2). All patients with metastatic disease developed liver metastases. Median time to distant metastasis development was 29.4 months (range 12.5–46.3). Actuarial 2- and 4-year rates of choroidal melanoma-specific survival were 97.0% (CI 80.4–99.6) and 92.8% (CI 73.9–98.2), and for overall survival were 97.0% (CI 80.4–99.6) and 88.9% (CI 69.1–96.3), respectively (Table 2, Fig. 2).

Visual acuity & radiation toxicity

Visual acuity outcomes are reported in Table 3. Patients were placed into one of four general vision acuity categories: ≥ 20/40 (LogMAR 0.30), 20/50 to 20/200 (LogMAR 0.40 to 1.00), < 20/200 (LogMAR <1.00) to hand movements, and light perception/no light perception. Compared to pre-treatment testing, 14.7% of patients had no change in their visual acuity following treatment, 38.2% had one category level decline in visual acuity, 35.3% had a two-level decline in visual acuity, and 11.8% had a three-level decline in visual acuity when evaluated at last follow-up. For the 14.7% who had no change in their visual acuity, the median follow-up time was 2.9 years (range 1.7–4.6). Radiation-related toxicity rates are listed in Table 3. For evaluable patients, optic atrophy was evident in 45.8% (11/24) of patients, while radiation retinopathy was seen in 87.5% (21/24). Neovascular glaucoma was observed in 17.7% (6/34) of patients. In addition to the one patient with local recurrence who had an enucleation, one additional patient had an enucleation due to eye pain from the development of glaucoma. No residual tumor was found in the enucleated

Table 2 Actuarial 2- and 4-Year Oncologic Outcomes

	Actuarial 2-Year (CI)	Actuarial 4-Year (CI)
Eye Preservation	96.9% (80.1–99.6)	96.9% (80.1–99.6)
Local Control	96.9% (80.1–99.6)	92.4% (72.3–98.1)
Distant Metastasis-Free Survival	97.0% (80.4–99.6)	88.5% (68.1–96.2)
Choroidal Melanoma-Specific Survival	97.0% (80.4–99.6)	92.8% (73.9–98.2)
Overall Survival	97.0% (80.4–99.6)	88.9% (69.1–96.3)

CI 95% confidence interval

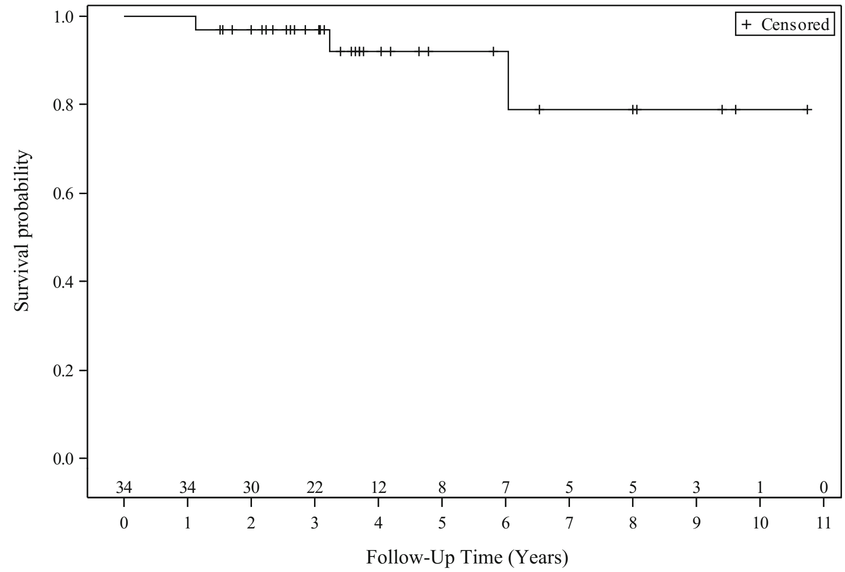

Fig. 1 Kaplan-Meier analysis of local control. Actuarial 2- and 4-year rates of local recurrence were 3.1% and 7.6%, respectively

pathologic specimen. Therefore, 94.1% (32/34) of patients retained their affected eye at last follow-up.

Discussion

In this series of exclusively juxtapapillary choroidal melanoma with 0 mm separation from the optic disc and at least partial encasement of it, encouraging eye preservation and local control rates exceeding 90% were seen following I-125 notched plaque brachytherapy at a 3.7-year median follow-up. In addition to these favorable local tumor control rates, patients also had largely acceptable visual outcomes, with 23.5% achieving 20/200 or better final vision.

When comparing visual acuity outcomes with the medium-sized COMS trial, where only about 15% of patients had peripapillary tumors, 34% had acuity ≥20/40, while 45% had acuity ≤20/200 at 3 years post-treatment [10]. While our visual acuity results are less favorable as compared to the COMS data, the COMS was a trial for which the patients in our report would not have been eligible given the 0 mm distance of the tumor from the optic disc for this cohort's patients. Additionally, with the somewhat longer follow-up of our report compared to the COMS, we expected a less favorable visual outcome.

Two other series are comparable to our patient cohort. Sagoo et al. [2] reported on 37 patients with circumpapillary lesions, all of which completely encircled the optic disc (360^0). Local recurrence occurred in 14%, and 25% required

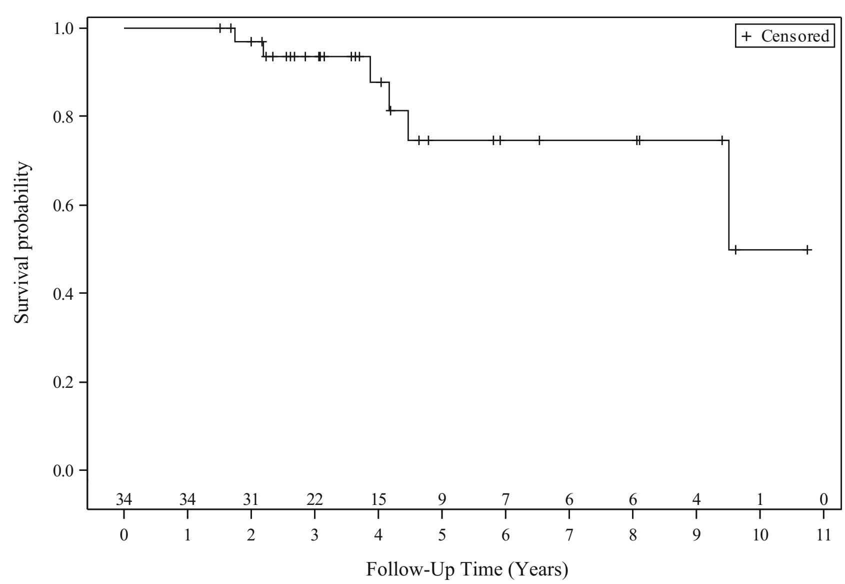

Fig. 2 Kaplan-Meier analysis of overall survival. Actuarial 2- and 4-year rates of overall survival were 97.0% and 88.9%, respectively

Table 3 Visual Acuity & Radiation Toxicities

Visual acuity <20/200 (< LogMAR 1.00):	
Prior to plaque placement	5/34 (14.7%)
At last follow-up	26/34 (76.5%)
Visual acuity prior to plaque placement	
≥ 20/40 (≥ LogMAR 0.30)	22/34 (64.7%)
20/50 to 20/200 (LogMAR 0.40 to 1.00)	7/34 (20.6%)
< 20/200 to hand movements (< LogMAR 1.00)	5/34 (14.7%)
Light perception or no light perception	0/34 (0)
Visual acuity at last follow-up	
≥ 20/40 (≥LogMAR 0.30)	3/34 (8.8%)
20/50 to 20/200 (LogMAR 0.40 to 1.00)	5/34 (14.7%)
< 20/200 to hand movements (<LogMAR 1.00)	17/34 (50.0%)
Light perception or no light perception	9/34 (26.5%)
Extent of vision decline from pre-treatment to last follow-up*	
No decline	5/34 (14.7%)
One-Level Decline	13/34 (38.2%)
Two-Level Decline	12/34 (35.3%)
Three-Level Decline	4/34 (11.8%)
Radiation Retinopathy	
No	3/24 (12.5%)
Yes	21/24 (87.5%)
Optic Atrophy	
No	13/24 (54.2%)
Yes	11/24 (45.8%)
Neovascular Glaucoma	
No	28/34 (82.3%)
Yes	6/34 (17.7%)

LogMAR Logarithm of the Minimum Angle of Resolution

*Level decline refers to the four grouping system presented in this table of (≥ 20/40, 20/50 to 20/200, <20/200 to hand movements, and light perception/no light perception)

enucleation. Sixty-two percent had final visual acuity ≤20/200 (with this value excluding the seven enucleated eyes). It is worth nothing that while the Sagoo et al. cohort had somewhat higher local recurrence and enucleation rates than our series, their entire cohort had tumor completely encircling of the optic disc, while our study included 44.1% with less than 50% circumferential involvement. However, treatments were far more heterogeneous in the Sagoo et al. series than our own. For the radiation treatment in Sagoo et al., three patients were treated with Cobalt-60, three patients were treated with a round-shaped rather than notched plaque, and the treatments had apical doses ranging from 45 to 125 Gy. In addition, 70% of patients received some planned adjunctive treatment including transpupillary thermotherapy or argon laser photocoagulation. In our series, the same radiation treatment (I-125 notched plaque brachytherapy to a dose of 85 Gy) was performed without adjuvant therapy for any patient. The one source of treatment heterogeneity in our own cohort was that 20% received silicone oil treatment during the plaque procedure for radiation attenuation.

In Finger et al. [3], 24 patients with juxtapapillary and circumpapillary choroidal melanoma were treated with Palladium-103 slotted eye plaque radiotherapy. With 1.9 years of follow-up, there was 100% local control. No enucleations were required, and 77% had visual acuity >20/200 at last follow-up. These outcomes are better than those reported in our current study. However, comparison of our study to the report by Finger et al. [3] is limited due to the inclusion of patients which did not meet our study criteria, as 13% of patients had lesions not even touching the optic disc. Additionally, only one patient (4%) had a T3 tumor (in contrast to 14.7% in our study). Also, Pd-103, rather than I-125, was utilized as the brachytherapy source. Lastly, the Finger et al. study had an approximately 50% shorter median follow-up time than our study, which may account at least partially for their study's better visual acuity results.

Given the limited data available on juxtapapillary and circumpapillary lesions, we compared our results to brachytherapy series which included juxtapapillary and circumpapillary tumors. A list of recent brachytherapy series of these tumor types is reported in Table 4. In these studies, 4- to 5-year local control rates range from 86 to 90% and visual acuity >20/200 ranges from 23 to 46%, which are similar to those reported in our study. In comparing our results to one of the largest series reported of 141 peripapillary choroidal melanomas patients, local failure occurred in 10% with a median follow-up of 46 months [14]. Of note, adjuvant transpupillary thermotherapy was utilized in 39% of these patients. In terms of visual acuity, 77% developed ≤20/200 vision, similar to our own study with a similar length of follow-up.

Proton beam and linear accelerator (LINAC)-based radiotherapy are other radiation options which can be used to treat choroidal melanoma. A summary of data using external beam radiation techniques is reported in Online Resource 1. The 4- to 5-year local control rates range from 86 to 93% in the proton and LINAC-based stereotactic radiotherapy (SRT) series, and visual acuity rates ≥20/200 range from 10 to 31.3% [13, 16–20]. It is difficult to directly compare our results to those of an external beam series because of the potential for case selection bias and different lengths of follow-up. Regardless, compared to these techniques, brachytherapy appears to demonstrate comparable levels of local tumor control and visual acuity rates.

Additionally, helium ion and carbon ion treatments are alternative options for choroidal melanoma. With long-term follow-up of a prospective study comparing helium ion therapy to I-125 plaque brachytherapy <2 mm from the optic disc, improved local control, eye preservation, and disease-free survival were seen with helium ion therapy [21]. While compelling, the brachytherapy technique used in the study (no ultrasound guidance and small treatment margin) may have contributed to the worse brachytherapy outcomes.

Table 4 Literature Review of Reported Series of Circumpapillary and Juxtapapillary Choroidal Melanoma Treated with Plaque Brachytherapy

Study	No. Patients	Tumor Location	Dose (Gy)	Follow-up Period	Eye Preservation	Local Control	DMFS	DSS	Toxicity Profile
Current Study	34	Circumpapillary/ Juxtapapillary	85	Actuarial 2 yr. Actuarial 4 yr	96.9% 96.9%	96.9% 92.4%	97.0% 88.5%	97.0% 92.8%	Visual acuity ≥20/200 in 23.5% Radiation retinopathy in 87.5%
Sagoo et al. [2]	37	Circumpapillary	80	3.8 yr	75%	86%	96%	100%	Visual acuity ≤20/200 in 62% Radiation retinopathy in 39%
Finger et al. [3]	24	Circumpapillary/ Juxtapapillary	85	1.9 yr	100%	100%	100%	–	Visual acuity >20/200 in 77% 41% lost ≥2 lines of vision
Sagoo et al. [11] Sagoo et al. [12]	650	Juxtapapillary	80	3.33 yr. Actuarial 2 yr. Actuarial 5 yr	– 94% 84%	– 93% 86%	– 98% 89%	– 99% 96%	– Visual acuity >20/200 in 82% Visual acuity >20/200 in 46% Non-proliferative (Proliferative) Radiation Retinopathy in 66% (24%)
Krema et al. [13]	30	Juxtapapillary	85	3.8 yr. Actuarial 4 yr	– –	– 89%	– –	– –	– Radiation retinopathy in 60%
Sagoo et al. [14]	141	Juxtapapillary	85	4.7 yr	77%	90%	87%	97%	Visual acuity >20/200 in 23% Non-proliferative (Proliferative) Radiation Retinopathy in 51% (22%)
De Potter et al. [15]	93	Juxtapapillary	85	6.5 yr. Actuarial 5 yr	78% –	85% –	88% –	– –	Radiation retinopathy in 87% Radiation retinopathy in 94%

DMFS distant metastasis-free survival, *DSS* disease-specific survival

Several limitations exist for this study. One limitation is that this is a small series of patients. This is also a retrospective study, and there may have been selection bias in recommending plaque brachytherapy over other alternatives for this patient cohort. The median follow-up of 3.7 years is comparable to other studies evaluating this subset of patients with choroidal melanoma. However, the median time for local relapse was 38.7 months, and 15 patients (44%) did not reach this follow-up period. Therefore, although Kaplan-Meier analysis was used to estimate oncologic outcomes, the lack of long-term follow-up in the entire cohort may have affected the results. Furthermore, 21% of the cohort had silicone oil utilized for radiation attenuation, which results in some treatment heterogeneity. This may have potentially affected oncologic and/or toxicity outcomes. Additionally, the guidelines for treatment planning and plaque placement were updated over the span the study cohort was treated, for which the implementation may improve local control rates. With improved local control, improved outcomes should be expected [22]. Lastly, not all patients were evaluable for radiation toxicities at follow-up given limited funduscopic examinations as a result of media opacities or tumor involvement of the macula.

In conclusion, we report our institution's 10-year experience with juxtapapillary choroidal melanoma directly contiguous with the optic disc and partially encircling it treated with ultrasound-guided, I-125 notched plaque brachytherapy. Our experience, albeit with a small cohort, demonstrates that a globe-sparing, I-125 brachytherapy technique is a feasible option for these cases. However, prospective studies generalizable to a larger population, which compare modern brachytherapy and external beam radiotherapy methods, in particular for patients with juxtapapillary lesions, may help elucidate the optimal treatment technique for this clinical scenario.

Compliance with ethical standards

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Conflict of interest Aside from the listed author affiliations and funding, the authors certify that they have no other affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

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